INTEGRATED APPLICATION ? FEATURE DELIVERY ? NOVEMBER 2026

Flood Risk & Disaster Management

Progress report | Implemented application features and model-training status

PREPARED BY	REPORTING PERIOD	STATUS
Flood Risk & Disaster Management	November 2026	Active development

REPORT SUMMARY

The system combines flood monitoring, GIS screening, incident/report workflows, data stewardship and scenario/result analysis. The points below describe features present in the supplied application code and research workspace. Model evidence and operational readiness are reported separately on page 2.

Implemented app features

- **Operational dashboard ?** Area selection, KPI/status panels, last-refresh time, provider error visibility, live-data refresh and a timeline of recent operational activity.
- **Weather & river monitoring ?** Open-Meteo rainfall, temperature, humidity and soil-moisture inputs; GloFAS modelled discharge, change and next-day context; station and hydrograph views. Modelled flow is kept distinct from measured river level.
- **Flood-risk screening ?** Rule-based station risk score, severity/status and contributing factors. A national weather-based GIS screening layer covers 126 Pakistan districts; station-specific river-aware predictions are available for configured monitoring locations.
- **GIS operations & map tools ?** District and river layers, rainfall rasters, glaciers, coordinates and catchment-focused exploration; layer toggles, district selection and map-based inspection. Boundaries and risk coverage are presented as separate data layers.
- **Forecast and analytics views ?** Station forecast cards and charts, historical/rainfall analytics, monthly outlook and longer-horizon exposure views. Trimmu/Balloki XGBoost forecast artifacts support 24/48-hour prototype outputs; the current station-training table is synthetic.
- **Citizen reports, incidents & alerts ?** Create/list and persist citizen reports; mark reports important or closed; track incident records, alert/event history and review/approval status. Dashboard activity and automation-context endpoints support follow-up.
- **Satellite water-extent analysis ?** Sentinel-1 VV/VH imagery with elevation/slope inputs produces a georeferenced water overlay and area/coverage summaries. A permanent-water reference can be removed to display potential flooding; historical examples support comparison with labels.
- **Scenario preparation & simulation ?** Scenario editor and JSON import/export; register and check terrain, rainfall, inflows and other inputs; prepare a domain, bind an engine project, queue computation and publish checked results. External HEC-RAS 2D and LISFLOOD-FP engines are supported.
- **Model-result review ?** Map playback of timestamped depth, water-surface and speed rasters; timeline, hydrograph, playback controls and A/B comparison of runs with timestamp offsets shown.
- **Data, exposure & persistence ?** Source/provenance and quality-review pages with discharge/rainfall plots; Marala dataset tools and a Trimmu/Jhang exposure study using population, roads, buildings, facilities, settlements and historic flood references. SQLite-backed records and versioned APIs support dashboard data.

FORECASTING MODELS ? TRAINING DATA ? EVIDENCE STATUS

Models trained & data used

Training, calibration and inference are different stages. The list below names completed model work and identifies what is currently suitable for operational use.

River-flow forecasting research

- **Models trained ?** Weather-only and past-discharge LSTMs, an encoder-decoder LSTM, a spatial elevation-band LSTM, N-BEATSx, N-HiTS, and HBV / delta-HBV physical and ML-residual variants (LSTM and LightGBM). These are research hindcasts; they are not deployed live forecasts.
- **Training observations ?** Daily Marala inflow from WAPDA, 1 March 2016?19 December 2024: 3,216 dates, with 3,209 passing current source-consistency checks and seven unresolved dates withheld. Training splits for 30-day experiments: 2016?2020 train, 2021 validation, 2022?2024 retrospective test.
- **Weather/forcing inputs ?** Historical weather and snow features, including CHIRPS v3 catchment-weighted rainfall where configured. The 30-day studies use noisy ERA5-Land reanalysis as a stand-in for future weather because archived 30-day forecasts were unavailable. Scores therefore do not establish operational forecast accuracy.

Flood-risk and station ML

- **Pakistan flood-risk XGBoost ?** A synthetic development regressor was trained on 6,000 generated rows and is disabled by default. Real-data classifiers used Global Flood Database event windows (28 Pakistan events, 2003?2018; 2,410 daily rows) plus Open-Meteo historical weather/soil moisture and GloFAS discharge. A richer DesInventar set used 10,097 district-date rows; a Lower Indus pilot used 4,597 rows. Geographic/temporal validation gates failed, so these research artifacts are not deployed.
- **Station discharge XGBoost ?** Trimmu +24-hour and Balloki +48-hour regression baseline uses current and lagged flow, rainfall windows, upstream discharge, release, soil moisture and forecast rainfall. Its bootstrap training data are synthetic; it is a development baseline, not a validated station forecast.

Satellite and hydrologic models

- **SAR U-Net water mapping ?** A trained U-Net checkpoint is packaged for Sentinel-1 water-extent inference. Sen1Floods11 evaluation records test IoU 65.36% and Dice 79.05%; Pakistan test IoU is 13.47%. Event-sharing splits do not prove unseen-event accuracy, and mapped water is not automatically new flooding.
- **HEC-HMS routing calibration ?** A HEC-HMS 4.14 Muskingum routing experiment used visually reviewed 2022 FFC daily gauge reports for Warsak, Swat/Charsadda Road and Nowshera. Seven Nowshera observations were used for fitting; eight later observations from the same event were held out. Withheld NSE/KGE was 0.678/0.462; the KGE target failed. This is fitted hydrologic routing, not a trained neural model.
- **2D hydraulic simulations ?** HEC-RAS and LISFLOOD-FP cases demonstrate external-engine preparation, computation and map review. Channel-bed geometry remains estimated and independent flood-extent validation is unavailable; published results are exploratory, not accepted calibrated forecasts.

Data priority: reviewed sub-daily gauge series, surveyed channel geometry, archived forecast weather and independent event flood extents are needed to train, calibrate and validate operational models.